

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 3 – NUMERICAL HACKATHON

Course Code:	ITA0303	Course Title:	Mobile computing
CO Assessed:	CO3 – Precision (BL3)	Assessment:	Assessment Tool 3 – HACKATHON
Weightage:	30 % of CIA	Total Marks:	100
CO3 Statement:	Implement wireless networking and Mobile IP concepts for routing in cellular and ad hoc networks.. (BTL 3)		
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Section / Batch:		Date:	

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Duration: 60 minutes
- Number of questions : 10

Faculty In-charge	Course Coordinator	Head of Department
Mrs. B. Abisha	Dr. Terrance Frederick Fernandez	Dr. Sasi Rekha
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. Mobile IP Tunnel Overhead

A logistics company uses **Mobile IP** to allow delivery vehicles to remain connected while moving between cities.

The communication between the **home agent** and **foreign agent** is established using a tunneling mechanism.

The tunneling process adds an overhead of **20 bytes per packet** during transmission.

One vehicle sends a **500-byte data packet** to the headquarters.

Calculate the additional overhead introduced by the tunnel for this packet.

2. Maximum Theoretical Data Rate

A university campus deploys a wireless network to support high-speed multimedia applications.

The network operates with a **bandwidth of 20 MHz** and uses **64-QAM modulation**.

Each transmitted symbol carries **6 bits** of information.

The administrator wants to estimate the network's maximum theoretical throughput.

Calculate the maximum theoretical data rate of the wireless network.

3. Routing Table Memory Requirement

A proactive routing protocol is implemented in a military communication network to ensure routes are always available.

Each mobile node maintains a routing table containing **50 entries**.

Every routing entry occupies **32 bytes** of memory.

The network administrator needs to estimate the memory consumed by the routing table.

Calculate the total memory required for storing the routing table.

4. Distance Travelled Between Routing Updates

An emergency response vehicle communicates through an **ad-hoc wireless network** while moving across a disaster zone.

The vehicle travels at a constant speed of **10 m/s**.

The routing protocol updates its routing information every **2 seconds**.

The system designer wants to know how far the vehicle moves before each update occurs.

Calculate the total distance traveled between consecutive routing updates.

5. DSR Route Discovery

A rescue operation uses the **Dynamic Source Routing (DSR)** protocol to establish communication between two mobile devices.

During route discovery, the route request packet passes through **5 intermediate nodes** before reaching the destination.

Each node forwards the request only once to continue the discovery process.

The network engineer wants to determine the number of route request packets involved in this process.

Calculate the total number of route request packets generated.

6. Mobile IP Subnet Addressing

A Mobile IP-based enterprise network is divided into multiple subnets for efficient address management.

One subnet is configured with the subnet mask **255.255.255.240**.

The network administrator plans to allocate IP addresses to mobile nodes within this subnet.

Before deployment, the maximum number of usable host addresses must be determined.

Calculate how many mobile nodes can be assigned IP addresses in one subnet.

7. Spectral Efficiency

An IoT-enabled smart factory operates its wireless communication system at **2.4 GHz**.

The communication channel has a **bandwidth of 20 MHz**.

The transmitter operates at a **symbol rate of 1 Msps**.

The engineer needs to evaluate how efficiently the available bandwidth is utilized.

Calculate the spectral efficiency of the wireless communication system.

8. Network Density

A wildlife monitoring project deploys an **ad-hoc wireless sensor network** across a protected forest area.

The network consists of **50 sensor nodes**, each having a **transmission range of 250 meters**.

The total deployment area is **1 km²**.

The project team wants to estimate the density of nodes within the network.

Calculate the approximate network density.

9. Doppler Shift

A high-speed train provides onboard wireless connectivity while traveling through a cellular network.

The train moves at a speed of **30 m/s**.

The wireless communication system operates at a carrier frequency of **900 MHz**.

The network engineer needs to evaluate the effect of motion on the received signal frequency.

Calculate the Doppler shift experienced by the mobile node.

10. Packet Transmission Time

A company uses a **wireless LAN** to transfer large engineering design files between workstations.

The WLAN supports a data rate of **54 Mbps**.

Each transmitted packet has a size of **1500 bytes**.

The network administrator wants to estimate the transmission delay for a single packet.

Calculate the transmission time required to transmit one packet.

Assessment tool 3:-

1) Mobile IP - Tunnel overhead:-

Given:-

$$\text{Data Packet} = 500 \text{ bytes}$$

$$\text{Tunnel overhead} = 20 \text{ bytes.}$$

$$\text{Additional overhead} = 20 \text{ bytes.}$$

Total transmitted size :

$$500 + 20 = 520 \text{ bytes.}$$

$\therefore$ 20 bytes overhead.

2) Maximum Theoretical Data Rate:-

Given:-

$$\text{Bandwidth} = 20 \text{ MHz} = 20 \times 10^6 \text{ Hz.}$$

$$64\text{-QAM} = 6 \text{ bits / Symbol.}$$

$$\text{Data Rate} = \text{Bandwidth} \times \text{bits Per Symbol.}$$

$$= 20 \times 10^6 \times 6.$$

$$= 120 \times 10^6.$$

$$\therefore 120 \text{ Mbps.}$$

3.) Routing table memory requirement :-

Given :-

$$\text{Entries} = 50.$$

$$\text{Each entry} = 32 \text{ bytes}.$$

$$50 \times 32 = 1600 \text{ bytes}.$$

$$1600 \text{ bytes} = 1.5625 \text{ KB}.$$

$$\therefore 1600 \text{ bytes } (\approx 1.56 \text{ KB}).$$

4.) Distance travelled Between routing updates :-

$$\text{Speed} = 10 \text{ m/s}.$$

$$\text{update interval} = 2 \text{ s}.$$

$$\text{Distance} = \text{Speed} \times \text{time}.$$

$$= 10 \times 2 = 20 \text{ m}.$$

5.) DSR Route Discovery :-

* 5 intermediate nodes.

* Each node forwards the request once.

* Source also generates the initial request.

Number of route request :-

$$1 + 5 = 6.$$

If counting only generated RREQ packets at the nodes.

$\therefore$ 6 route request packet transmission.

6.) Mobile IP Subnet addressing:-

Subnet mask :-

$$255.255.255.240.$$

This is /28.

Number of host bits :-

$$32 - 28 = 4.$$

Total addresses :-

$$2^4 = 16.$$

usable host addresses :-

$$16 - 2 = 14.$$

$\therefore$ 14 mobile nodes.

7.) Spectral efficiency:-

$$\text{Symbol rate} = 1 \text{ Msps} = 1 \times 10^6 \text{ Symbol / s.}$$

$$\text{Bandwidth} = 20 \text{ MHz} = 20 \times 10^6 \text{ Hz}.$$

Data rate :-

$$1 \times 10^6 \text{ bit/s} = 1 \text{ Mbps}.$$

Spectral efficiency :-

$$\eta = \frac{\text{Data rate}}{\text{Bandwidth}}.$$

$$= \frac{1}{20} = 0.05 \text{ bit/s/Hz}.$$

8.) Network density :-

$$\text{Number of nodes} = 50.$$

$$\text{Area} = 1 \text{ km}^2.$$

$$\text{Density} = \frac{\text{Number of nodes}}{\text{Area}}.$$

$$= \frac{50}{1} = 50.$$

$$\therefore 50 \text{ nodes/km}^2.$$

9.) Doppler shift :-

$$\text{Speed } v = 30 \text{ m/s}.$$

$$\text{carrier frequency} = 900 \text{ MHz}.$$

Speed of light $c = 3 \times 10^8 \text{ m/s}$.

$$f_d = \frac{v f_c}{c}$$

$$900 \text{ MHz} = 900 \times 10^6 \text{ Hz}$$

$$f_d = \frac{30 (900 \times 10^6)}{3 \times 10^8}$$

$$= 90 \times 10^1$$

$$= 9000 \text{ Hz}$$

10.) Packet transmission time:-

$$\text{Data rate} = 54 \text{ Mbps}$$

$$\text{Packet size} = 1500 \text{ bytes}$$

convert bytes to bits

$$1500 \times 8 = 12000 \text{ bits}$$

$$\text{Transmission Time} = \frac{\text{Packet size}}{\text{Data rate}}$$

$$= \frac{12000}{54 \times 10^6} = 0.0002222 \text{ s}$$

$$0.0002222 \times 1000 = 0.2222 \text{ ms}$$